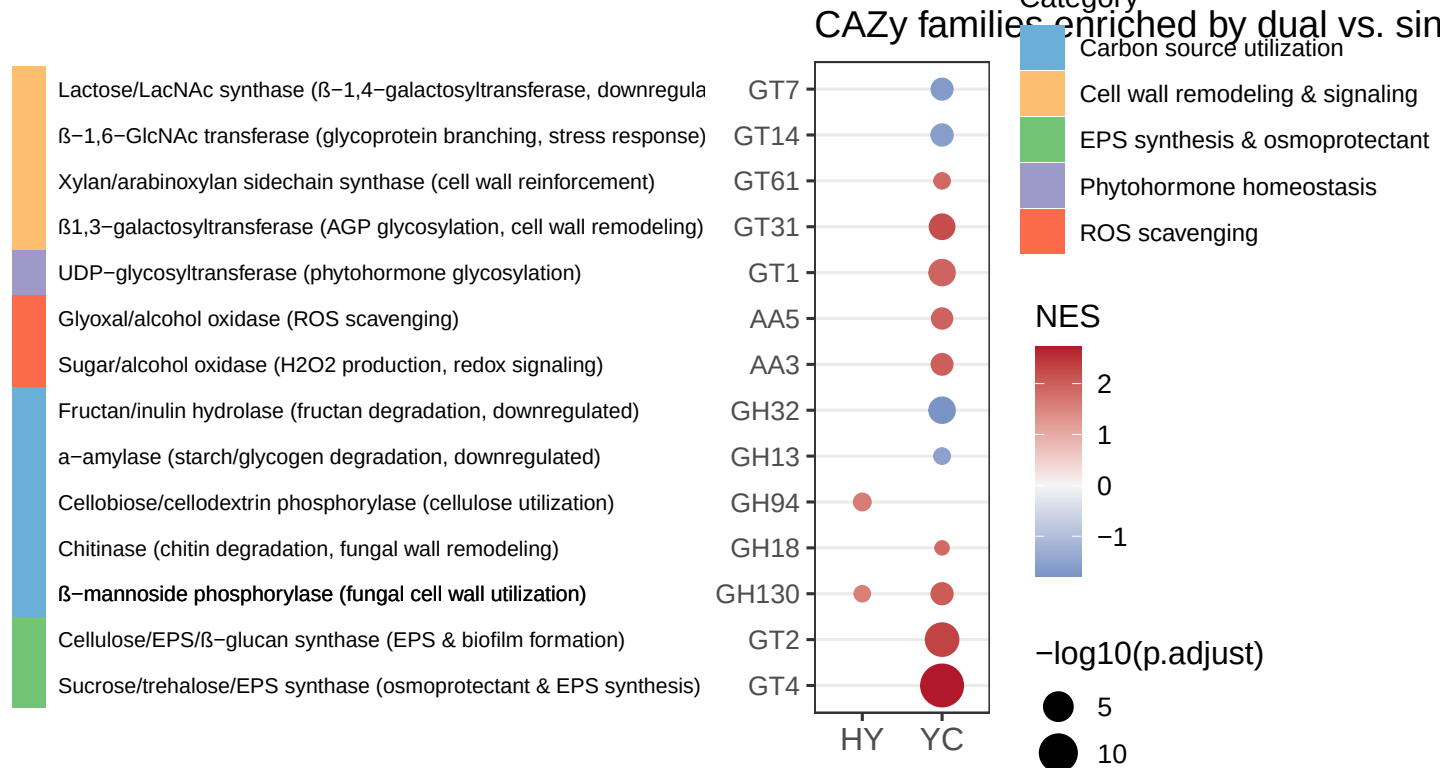
CAZy families enriched by dual vs. sin
Category
Carbon source utilization
Cell wall remodeling & signaling
EPS synthesis & osmoprotectant
Phytohormone homeostasis
ROS scavenging
Lactose/LacNAc synthase (ß−1,4−galactosyltransferase, downregula
ß−1,6−GlcNAc transferase (glycoprotein branching, stress response)
Xylan/arabinoxylan sidechain synthase (cell wall reinforcement)
ß1,3−galactosyltransferase (AGP glycosylation, cell wall remodeling)
UDP−glycosyltransferase (phytohormone glycosylation)
Glyoxal/alcohol oxidase (ROS scavenging)
Sugar/alcohol oxidase (H2O2 production, redox signaling)
Fructan/inulin hydrolase (fructan degradation, downregulated)
a−amylase (starch/glycogen degradation, downregulated)
Cellobiose/cellodextrin phosphorylase (cellulose utilization)
Chitinase (chitin degradation, fungal wall remodeling)
ß−mannoside phosphorylase (fungal cell wall utilization)
Cellulose/EPS/ß−glucan synthase (EPS & biofilm formation)
Sucrose/trehalose/EPS synthase (osmoprotectant & EPS synthesis)
GT7
GT14
GT61
GT31
GT1
AA5
AA3
GH32
GH13
GH94
GH18
GH130
GT2
GT4
HY
YC
NES
2
1
0
−1
−log10(p.adjust)
5
10